

ASTHMA EDUCATION

Metered-Dose Inhaler Technique: The Effect of Two Educational Interventions Delivered in Community Pharmacy Over Time

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Instruction is critical in order to ensure correct technique with pressurized metered-dose inhalers (pMDIs) by patients. The aim of this study was to compare the effects over time of two educational interventions delivered in community pharmacy to pMDI users. In this randomized controlled parallel-group study, pMDI technique was assessed before and after written and verbal instruction, alone or with physical demonstration, at baseline and 4, 8, and 16 weeks. The study recruited 52 subjects with asthma or chronic obstructive pulmonary disease (COPD). Initially only 1/52 (6%) subject had correct pMDI technique (= checklist score 8/8), with mean baseline score 5 (*SD* 1) for both groups. Written and verbal information improved pMDI technique at 16 weeks (7 ± 1 , $p < .05$). Addition of physical demonstration resulted in significant improvement at weeks 4, 8, and 16 (7 ± 1 , 7 ± 1 , 7 ± 1 respectively; $p < .05$ for each). Subjects receiving written and verbal information alone were less likely to return for follow-up than those receiving physical demonstration (8 weeks: 6/25 versus 19/27; $p < .001$). By the 8-week visit, 80% subjects in the physical demonstration group had correct technique prior to education, compared with 10% of subjects receiving written and verbal information alone ($p < .05$). There was some decline in inhaler technique by 16 weeks. The results demonstrate that adding a physical demonstration is more effective in improving pMDI technique than written and verbal instructions alone.

Keywords community pharmacy, demonstration, inhaler technique, instructions, metered-dose inhaler

INTRODUCTION

It is accepted that the optimal method for drug administration in respiratory conditions such as asthma and chronic obstructive pulmonary disease (COPD) is via inhalation (1). Of the range of inhalation devices available, the pressurized metered-dose inhaler (pMDI) continues to be the most commonly used device. It is the least expensive (2), least time-consuming to use, and more portable and efficient than a nebulizer (3, 4). However, many patients (5–7, 8) and health care providers (9–11) are not able to demonstrate correct pMDI technique.

Correct inhalation technique is crucial to optimal drug delivery to the airways for both controller and reliever medications. For β_2 -agonists, incorrect technique may lead to increased frequency of inhaler use (12, 13), which is an indicator of poorly controlled asthma, whereas incorrect inhaler technique with controller medications such as inhaled corticosteroids may lead directly to poor disease control (14), as demonstrated by increased frequency of emergency department and health care services use (4, 14).

As a result, patient education about pMDI technique is very important in the management of asthma. Previous studies have shown that inhaler technique can be improved with education (15, 16) but deteriorates over time (17–19), implying that educational interventions may need to be repeated. Various methods of education about inhaler technique have

been studied in a wide range of clinical settings. Results indicate that group education is most effective, followed by video instruction and then personal instruction having been found to be most effective, in that order (7, 15, 16, 20).

Pharmacists have a critical role to play in the administration of medications. They are responsible for dispensing the medication and personally handing it over to the patient. This interaction provides a timely opportunity to instruct patients on the use and administration of inhaled medications. However, the role of the community pharmacist in delivering a brief and targeted educational interventions about pMDI technique has not previously been investigated. The objective of this study was to evaluate the effects of two methods of educational intervention, delivered in a community pharmacy setting, on pMDI technique, and to determine whether continued instruction is needed to maintain correct pMDI technique.

METHODS

Study Design

This study had a randomized, parallel-group single-blind design, with four patient visits to a community pharmacy. The assessments and interventions were conducted by two pharmacy student researchers, who were trained in the use and assessment of the pMDI. Prior to assessment of participants, the pharmacy student researchers independently assessed and compared the technique of a sample of pMDI users to ensure consistency of evaluation.

Subjects were recruited from a convenience sample of eight community pharmacies in Sydney, Australia. The study was approved by the University of Sydney Ethics Committee

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TABLE 1. —pMDI technique checklist, published by van der Palen and colleagues (6).

Step 1. Shake the inhaler
Step 2. Hold inhaler upright
Step 3. Exhale to residual volume
Step 4. Keep head upright
Step 5. Mouthpiece between teeth and lips
Step 6. Inhale slowly and press canister
Step 7. Continue slow and deep inhalation
Step 8. Hold breath for 5 seconds

and written informed consent was obtained from the pharmacists and patients. Patients over the age of 18 years, currently using a pMDI for asthma or COPD, were eligible to participate. Exclusion criteria were first-time pMDI users, subjects who did not self-administer their pMDI, or who could not speak or understood English, or who used a spacer.

Pharmacists identified eligible subjects as they entered the pharmacy with a request for a pMDI product, either by prescription or over-the-counter,¹ i.e., pharmacists identified consecutive customers. Customers who agreed were referred to the researcher, who conducted the study assessments and interventions out of direct sight and sound of the pharmacist. Neither the subjects nor the pharmacists were informed that pMDI technique instruction was the focus of the study; the aim of the study was presented as “extra counselling to people with asthma or chronic obstructive pulmonary disease who were using at least one pMDI product.” Randomization was performed by means of computer generated random group allocation, prior to study commencement.

Study Visits

At Visit 1 (baseline), data were collected regarding subject demographics, disease history, use of respiratory medications, and previous pMDI instruction. Subjects were then asked, “Can you show me how you would usually use your pMDI?” Inhaler technique was scored using a published 8-point checklist (Table 1) (6), with 1 point given for each step performed correctly (maximum score = 8, “correct technique”). Following assessment, the randomized educational intervention was delivered. One group received verbal instructions with written information (standard instruction group) and the other received verbal instructions and written information plus physical demonstration with a placebo pMDI (extended instruction group, see below). Subjects were then asked “Can you show me how you would usually use your pMDI?” using their own inhaler. The instruction and technique assessment were repeated until the subject demonstrated correct technique, or for a maximum of 3 times.

At Visit 2 (4 weeks $\pm$ 7 days after Visit 1) and Visit 3 (4 weeks $\pm$ 7 days after Visit 2), inhaler technique was reassessed. If pMDI technique was incorrect, the randomized instruction and technique assessment were repeated until correct technique was achieved, or for a maximum of 3 times. At each visit, the randomized instruction was not given if the subject demonstrated correct pMDI technique.

At Visit 4 (8 weeks $\pm$ 7 days after Visit 3), inhaler technique was reassessed. All subjects then received written and verbal instructions with physical demonstration.

pMDI Instruction

Standard instruction consisted of written and verbal instruction. The written instruction was the product information leaflet, packaged with the subject’s newly dispensed inhaler; this was returned to the patient to take home at the end of the visit. The verbal instruction consisted of the researcher reading through all steps of pMDI administration technique from the Product Information Leaflet with the patient, using the illustrations in the leaflet as a visual guide.

Extended instruction consisted of the written and verbal instruction outlined above, plus a physical demonstration, which was performed by the researcher, using a placebo pMDI.

In all cases, subjects used their own pMDIs to demonstrate inhaler technique.

Statistical Analysis

Based on a 10% difference in the proportion of subjects able to demonstrate correct pMDI technique before and after training (7), with power of .8 and $p = .05$ (6), a total of 10 subjects was required in each group. Mean pMDI score and the number of repetitions of instructions delivered were compared between groups at each visit using the Kruskal-Wallis test. Mean pMDI scores for Visits 1–3 and Visits 1–4 were compared within groups using the Friedman test. Statistical analyses were performed using SPSS (Statistical Package for the Social Science Inc. 2001, version 11). A significance level of .05 was used.

RESULTS

Subject Demographics

One hundred and five subjects were approached and 52 (19 males and 33 females) agreed to participate and were enrolled, with 26 being randomized to standard instruction and 26 to extended instruction. The number of patients returning at Visits 2–4 was 16 versus 23 (Visit 2), 6 versus 19 (Visit 3), and 6 versus 18 (Visit 4) for the standard instruction and extended instruction groups, respectively (Figure 1). There were significantly more drop-outs in the standard instruction group by Visit 3 ($\chi^2 = 11.2$, $p = .0008$). However, there was no trend towards lower or higher inhaler technique scores associated with early withdrawal from the study. Table 2 reports demographic data and medication use from Visit 1. There were no clinically important differences between the standard and extended instruction groups.

Previous pMDI Education

Eighty-three percent of subjects (45/52) reported that they had received pMDI technique instructions from a health care professional (Figure 2). For most subjects (62%), this only occurred at the time of first prescription, which was a mean of 13 years earlier. The instructions were most commonly reported to have been given by the general practitioner (33%), or by both the pharmacist and another health care practitioner (general practitioner/nurse/specialist, 25%). The majority of subjects (69%) had never had their pMDI technique checked;

¹ In Australia, short-acting β_2 -agonist inhalers are available from pharmacies without a prescription.

TABLE 2. —Subject characteristics.

	Standard instruction group (n = 25)	Extended instruction group (n = 27)	All subjects (n = 52)
Age (years), mean (SD)	43 (18)	47 (20)	45 (19)
Range	22–82 years	21–77 years	21–82 years
Duration of disease (years) (SD)	19 (13)	16 (13)	17 (13)
Range	1 month–45 years	1 month–49 years	1 month–49 years
Duration of pMDI use (years) (SD)	14 (11)	11 (7)	13 (9)
Range	1 month–38 years	1 month–25 years	1 month–38 years
Reported indication for pMDI use			
Asthma	80%	89%	85%
COPD	16%	11%	13%
Sleep apnea	4%	—	2%
Medication administered through pMDI			
Reliever	92%	92%	92%
ICS	25%	38%	32%
Combination ICS/LABA	4%	23%	14%

Note. ICS = inhaled corticosteroid; ICS/LABA = inhaled corticosteroid/long-acting β_2 -agonist.

21% said that their technique had been checked by a general practitioner, 6% by a specialist, and 4% by a pharmacist.

Baseline pMDI Technique

At Visit 1, prior to instruction, only 6% (3/52) subjects had correct pMDI technique (Figure 3), with mean scores being 5 ± 1 for both groups ($p > .05$; Figure 4). The most frequent errors were with exhaling before actuation (Step 3; 40% and 30% incorrect in standard and extended instruction groups, respectively), continuing slow and deep inhalation (Step 7; 16% and 33% incorrect in standard and extended instruction groups, respectively), and holding breath for 5 seconds (Step 8; 32% and 26% incorrect in standard and extended instruction groups, respectively).

Effect of Method of Instruction—Visit 1

At Visit 1, there was no significant difference in the number of times each instruction was needed for the standard and extended instruction groups (1.9 ± 0.9 and 1.6 ± 0.8 respectively; independent samples t test, $p > .05$). After instruction, there was a significant improvement in inhaler technique score for both groups, to 8 ± 1 and 8 ± 0 , respectively (two-way repeated measures analysis of variance [ANOVA], $p < .05$ for both).

Effect of Method of Instruction Over Time

Inhaler technique score, assessed at each visit prior to delivery of the educational intervention, improved significantly through the study in both randomization groups (Figure 3). In the standard instruction group, the increase in score from baseline was only statistically significant at Visit 4, when only 6 subjects returned for review ($p < .05$; Friedman test). In the extended instruction group, the increase in score from baseline was significant at Visits 2, 3, and 4 ($p < .05$ for each; Friedman test). Significant differences were seen between groups at Visits 2 and 3 but not at Visit 4 (Figure 3).

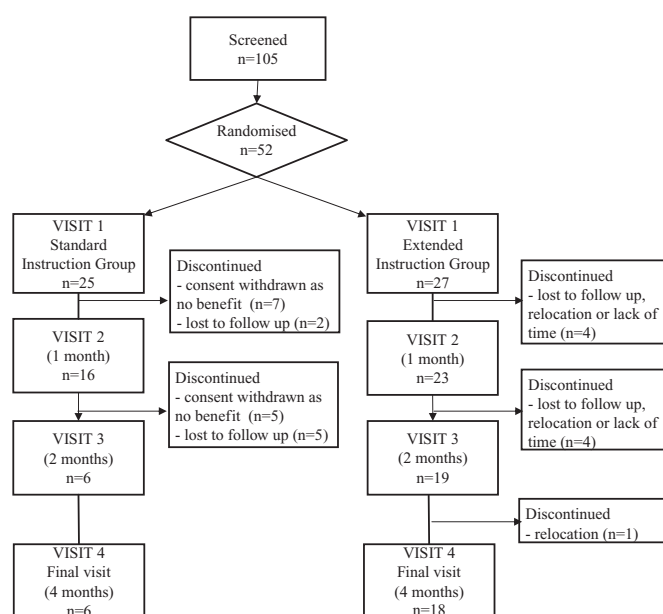


FIGURE 1.—CONSORT diagram showing flow of patients through the study. Standard instruction group received verbal and written information about inhaler technique; extended instruction group received verbal and written information plus physical demonstration with a placebo inhaler.

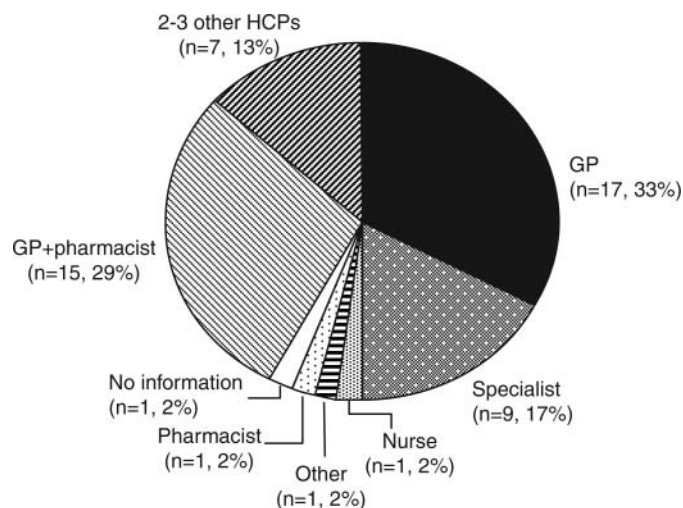


FIGURE 2.—Previous instruction about inhaler use (n = 52). HCP = health care professional (GP, pharmacist, specialist, nurse).

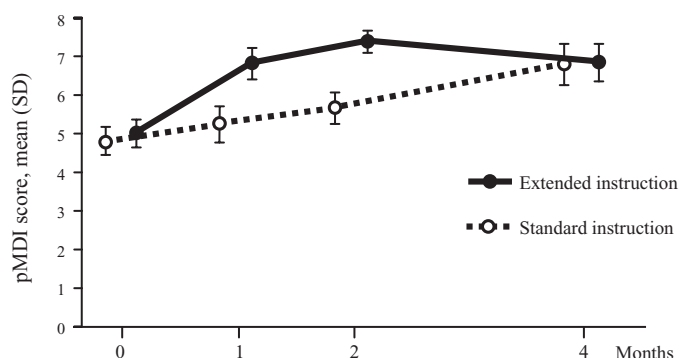


FIGURE 3.—pMDI technique score at each visit prior to instruction, $p < 0.05$, Kruskal-Wallis (one-way between group ANOVA) test.

In the standard instruction group, correct technique could not be demonstrated despite three instructions by 6/25, 4/16, and 3/6 subjects at Visits 1, 2, and 3, respectively. In the extended instruction group, only 1 subject was unable to demonstrate correct technique despite three instructions at each of Visits 1, 2, and 3 (attended by 27, 23, and 19 subjects respectively).

Over follow-up visits, there was a significant improvement in the percentage of patients demonstrating Step 8 correctly in the standard instruction group and Steps 3, 4, 7, and 8 correctly in the extended instruction group (data not shown).

Need for pMDI Instruction

There was no significant difference in the number of instructions that needed to be given during Visits 1, 2, and 3 in order to achieve correct technique between the standard and the extended counseling groups; however, there was a trend to fewer numbers of instructions being required to achieve correct technique in the extended instruction group (medians, 1, 1, and 0) compared with the standard instruction group (medians, 2, 2, and 1) for Visits 1, 2, and 3 (respectively).

DISCUSSION

This study is the first to investigate the effect of repeated educational interventions, delivered in a community pharmacy setting, on pMDI technique. The results demonstrate that adding a physical demonstration to verbal and written instructions is much more effective in improving pMDI technique compared with written and verbal instructions alone.

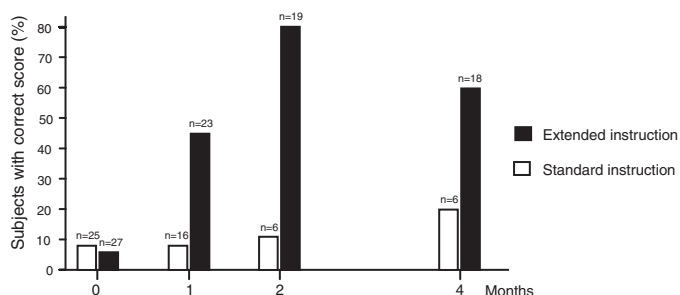


FIGURE 4.—Proportion of subject with correct pMDI technique score (8/8) at each visit.

The decline in technique after a subsequent 2-month gap suggests that repeated instruction is required to maintain correct inhaler technique, even in those individuals who had been using the devices for many years and have temporarily mastered the technique.

In this study, two forms of education were compared: verbal and written instruction (standard) versus verbal and written instruction with a physical demonstration with a placebo inhaler (extended). Of the methods evaluated in other clinical settings, these were selected as being most appropriate for a community pharmacy setting (7). Verbal instruction with written information was used as the reference intervention, as this was assumed to be the minimum standard a pharmacist should practise and is known to be the minimum level of instruction able to produce an improvement in pMDI technique (7). For ethical reasons, all patients were given optimum (extended) instruction at the end of the study.

The observation that a low proportion of pMDI users had correct inhaler technique at baseline (only 1 of 52 patients) is consistent with findings in other studies of pMDIs (1, 21–23) and as well as studies of technique with dry powder inhalers (20, 24). At this point, it is important to mention that although there is evidence indicating that there is more than one correct way to use a pMDI, the checklist used to evaluate pMDI technique in this study (6) is consistent with that the published literature (25).

When it comes to the possible causes of poor technique at baseline, the data collected on previous technique education are relevant. Although most patients reported having received some instruction about pMDI technique, this was mostly at the time of first prescription, and fewer than one third of patients had previously had their pMDI technique checked by a health professional. Of the health care professionals involved in the chain of steps between prescription of a medication and its effective use by a patient, pharmacists are in a pivotal position, as they are the most frequently and regularly seen by patients. Pharmacists have the ideal opportunity to check device technique during dispensing visits. If they were able to identify patients with incorrect inhaler technique, they could provide appropriate advice to such patients or refer them to their doctors, for example for consideration of a different type of inhaler or use of a spacer device. This study emphasizes the role the community pharmacist can play by not only correcting patients' pMDI technique but also by helping them maintain their technique over time.

This study is unique because it involved the assessment and education of pMDI technique over time. In this study, subjects underwent assessment and education at four pharmacy visits over 4 months. The first three visits were separated from each other by 1 month. This time frame was based on the typical interval between dispensing for a pMDI device (23). Visit 4 was conducted 2 months after the third visit in order to see whether correct pMDI technique could be sustained over a longer period. This study indicated that although there was a continuing improvement in pMDI technique when subjects were assessed and educated at monthly intervals, there was a decline in technique when no education was provided for 2 months.

Results from this study indicate that despite an improvement in technique in both groups, the extended education involving a physical demonstration with a placebo inhaler

was significantly more effective. The tool used to assess technique in this study was a previously published 8-step pMDI technique checklist (6). The difference in outcomes between the standard and extended interventions may be due to the fact that some of the steps involved in correctly using a pMDI, such as “continue slow and deep inhalation” (step 7), require the development of skills that may be easier for patients to achieve when they physically see the technique being demonstrated rather than just read and hear about them. This is consistent with the fact that technique improvement was mainly seen in Steps 3, 4, 7 and 8, the steps which were most often incorrect at baseline, and which mainly involve skill development. The results in this longitudinal study help to explain the previous work of van der Palen et al. (7), Self et al. (22), and Lindgren et al. (13), who found that even with one-off device instructions, a physical demonstration was required to demonstrate improvement in device technique.

From the patient's perspective, the impact of physical demonstration is perhaps reflected in the difference in discontinuations between the two groups. By the end of the study, only 6 subjects remained in the standard instruction group compared with 16 subjects in the extended instruction group. When questioned about their reasons for withdrawal, several patients in the standard instruction group commented that they were not seeing any benefits or improvements from the study. The Visit 4 results for this group may thus have been biased by selective return of subjects who perceived the intervention to have been useful. None of the extended instruction group gave ‘not seeing any benefits’ as a reason for withdrawal from the study; withdrawals from this group were due to moving away from the area, loss of contact, lack of time, etc.

The results of this study need to be repeated in a larger sample size, although it has confirmed the feasibility of delivering effective inhaler technique education within the physical confines of a community pharmacy setting (21), and has demonstrated the improved effectiveness obtained by using a physical demonstration of inhaler technique. The study is limited by the fact that randomization allocation was not concealed from the researcher, and interventions and assessments were carried out by the same researcher. However, patients were approached about the study by the pharmacist rather than the researcher, and the potential for bias in assessment of inhaler technique was limited by the use of a standard checklist with which the researcher assessed inhaler technique. We cannot exclude the possibility that outside advice was given by other health care professionals during the course of the study, but, if anything, this should have served to reduce rather than increase the difference between randomization groups.

From our results, we can conclude that both standard and extended instructions result in improved pMDI technique, with written information and verbal instruction plus a physical demonstration being the superior mode of instruction when delivered in the community pharmacy. Studies indicate that community pharmacists in practice are not currently using physical demonstration as a mean of instruction when instructing patients (9, 24). Furthermore, this study supports the need for regular review in asthma management, particularly when it comes to pMDI technique. Future studies should be conducted over a longer duration of time, with a larger sample size in order to determine the clinical benefits of

improving pMDI technique but also with the view to understand how improving device technique can impact on other facet of asthma management such as self-management and medication adherence.

DECLARATION OF INTEREST

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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